sional correlation. In this case, 'Y' is larger than 'X.'

2. 'same' (default) returns the central part of the correlation. In this case, 'Y' is of the same size as 'X.'

3. 'valid' returns only those parts of the correlation that are computed without zero-padded edges. In this case, 'Y' is smaller than 'X.'

There is no single best approach to use the filter2 function. However, the method is dictated by the problem at hand.

One can create filter by hand or by using the fspecial function. fspecial is an IPT function designed to simplify the creation of common 2D image filters. Its syntax is:

```
h = fspecial(type, parameters)
```

where:

1. 'h' is the filter mask
2. 'type' is one of these: 'average'—averaging filter, 'disk'—circular averaging filter, 'gaussian'—Gaussian low-pass filter, 'laplacian'—2D Laplacian operator, 'log'—Laplacian of Gaussian (LoG) filter, 'motion'—approximates the linear motion of a camera, 'prewitt' and 'sobel'—horizontal edge-emphasising filters, 'unsharp'—unsharp contrast-enhancement filter)
3. 'parameters' are optional and vary depending on the type of filter; for example, mask size, standard deviation (for 'gaussian' filter) and so on

imfilter is another command for implementing linear filters in MATLAB.

The syntax for imfilter is:

```
g = imfilter(f, h, mode, boundary_
options, size_options);
```

Where 'f' is the input image, 'h' is the filter mask, and 'mode' can be either 'conv' or 'corr,' indicating whether filtering will be done using convolution or correlation (which is the default), respectively.

boundary_options refer to how the filtering algorithm should treat border values. There are four possibilities:

Fig. 1: Original image

Fig. 2: Image using averaging filter of dimensions 3x3

Fig. 3: Image using an average filter of 9x9

Fig. 4: Image with salt and pepper noise added

1. X. The boundaries of the input array (image) are extended by padding with a value 'X.' This is the default option (with X=0).

2. 'symmetric'. The boundaries of the input array (image) are extended by mirror-reflecting the image across its border.

3. 'replicate'. The boundaries of the input array (image) are extended by replicating the values nearest to the image border.

4. 'circular'. The boundaries of the input array (image) are extended by implicitly assuming the input array is periodic, that is, treating the image as one period of a 2D periodic function.

As regards size_options, there are two size options for the resulting image: 'full' (output image is the full filtered result, that is, the size of the extended/padded image) or 'same' (output image is of the same size as input image), which is the default.

'g' is the output image.

Correlation. Correlation is the same as convolution without mirroring (flipping) of the mask before the sums of products are computed. The difference between using correlation and convolution in 2D neighbourhood processing operations is often irrelevant because many popular masks used in image processing are symmetrical around the origin.

Linear low-pass filters

The averaging filter is a linear LPF implemented using 'average' option in the fspecial function. The command given below produces an averaging filter of size 5x7:

```
fspecial('average', [5,7])
```

The output of this command in MATLAB is:

```
ans =

    0.0286    0.0286    0.0286    0.0286    0.0286    0.0286    0.0286
    0.0286    0.0286    0.0286    0.0286    0.0286    0.0286    0.0286
    0.0286    0.0286    0.0286    0.0286    0.0286    0.0286    0.0286
    0.0286    0.0286    0.0286    0.0286    0.0286    0.0286    0.0286
    0.0286    0.0286    0.0286    0.0286    0.0286    0.0286    0.0286
```

The code given below applies an averaging filter of dimensions 3x3 to